

JUSTIFICATION AND APPROVAL OTHER THAN FULL AND OPEN COMPETITION

1. Identification of the agency and the contracting activity.

This justification is executed by the Naval Medical Research Command (NMRC), Silver Spring, MD, on behalf of the Navy to meet Navy Expeditionary Medical Research requirements.

2. Nature and/or description of the action.

This procurement action supports the previous acquisition of the Dräger AviAir system, a flight-certified, in-mask physiological monitoring solution developed under a Cooperative Research and Development Agreement (CRADA) with US Naval Air Systems Command (NAVAIR). The AviAir system is designed to monitor pilot respiration and environmental conditions in real time, enabling early detection of physiological episodes such as hypoxia and hypercapnia. The system has been flight-tested and validated on Navy platforms including the T-45, F-18, and F-35.

3. Description of Supplies or Services.

This is a sole source acquisition due to the proprietary nature of the AviAir technology, which integrates Dräger's core medical-grade gas sensors and algorithms into a miniaturized, aircrew-worn device. The planned advancements for the Dräger AviAir system, including the integration of inhalation and exhalation flow sensing, will be achieved through a systematic approach involving detailed modeling, rigorous testing, and iterative refinement. Initially, advanced computational modeling will be used to simulate the in-mask flow dynamics, optimizing sensor placement and functionality. This will be followed by prototyping, where physical models will be created to test the accuracy and reliability of flow measurements. Laboratory testing will ensure that the components perform effectively under various flight conditions, maintaining precision and durability. Field trials will then validate the system in real-world scenarios, providing feedback for further refinement.

4. Identification of the statutory exceptions permitting other than full and open competition.

The statutory exception permitting other than full and open competition for this acquisition is

10 U.S.C. 2304(c)(1) and FAR 6.302-1 sole source (including brand name) acquisitions or FAR 13.106-1(b)(1) soliciting from a single source.

Identification of the statutory exceptions permitting other than full and open competition (Brand Name).

The statutory exception permitting other than full and open competition for this acquisition is 10 U.S.C. 2304(c)(1) or 10 U.S.C. 2304(c)(3) and FAR 6.302-1(c) for brand name justification.

5. Demonstration that the proposed contractor's unique qualifications or the nature of the acquisition requires use of the authority cited.

Dräger is the only vendor capable of providing the AviAir system. The technology is not commercially available and was developed specifically for military aviation under a CRADA in collaboration with NAVAIR under NSCRADA-NAWCADPAX-20-392. AviAir uses proprietary Dräger medical-grade sensors and algorithms, miniaturized for flight use. The technology and form of the device is protected by patents US 12292429 B2 and US12478808 B2. It is the only system with validated performance in altitude chambers and in-flight trials, and the only one approved for use on Navy platforms. No other vendor offers a comparable solution with the same required level of hardware, firmware, and support.

6. Description of efforts made to ensure that offers are solicited from as many potential sources as practicable.

Market research was performed, the results of which are in paragraph 8. This requirement will be posted to the government point of entry in accordance with FAR 5.202.

7. Determination by the Contracting Officer that the anticipated cost to the Government will be fair and reasonable.

The Contracting Officer shall determine the anticipated cost to the Government of the supplies/services covered by this Justification and Approval will be fair and reasonable by conducting a price analysis before an award is made. In addition, the Government anticipates that the negotiated contract price will reflect a discount from the vendor's commercial price list and will be determined fair and reasonable.

8. Description of the market survey conducted and the results, or a statement of the reasons a market survey was not conducted.

Market research was not conducted because Dräger is the only system, maintenance, and service provider for the AviAir. The technology is not commercially available and was developed specifically for military aviation under a CRADA in collaboration with NAVAIR under NSCRADA-NAWCADPAX-20-392.

9. Other facts supporting the use of other than full and open competition.

Not applicable

10. Listing of Sources, if any, which expressed, in writing, an interest in the acquisition.

Not applicable

11. Statement of actions, if any, the agency may take to remove or overcome any barriers to competition before any subsequent acquisition for the supplies or services required.

For the reasons set forth in Paragraph 5, (contracting activity) has no plans at this time to compete future contacts for the types of supplies/services covered by this document. If another potential source emerges, (NMRC) will assess whether competition for future requirements are feasible.

12. Point of Contact:

The point of contact at the Naval Medical Research Command, Silver Spring, MD, regarding the contents of the sole source justification are the Contracting Officers in the Material Management Department (MMD) and can be reached by emailing usn.detrick.nmrc.mbx.mmd-contracts@health.mil.

TECHNICAL AND REQUIREMENTS CERTIFICATION REQUIRED BY NAVSUP 4200.83 series

I certify that the facts and representations under my cognizance which are included in this justification and approval and which form a basis for this justification are complete and accurate.

Technical POC

WARNER.STEPHANIE.ANN.1409469210	Digitally signed by WARNER.STEPHANIE.ANN.1 409469210 Date: 2026.03.04 13:51:46 -05'00'	937-938-3917	03/04/2026
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Department Head

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Title	Department Head		
Department/Command	EP/NAMRU-D		

CONTRACTING OFFICER CERTIFICATION

I certify that this justification is accurate and complete to the best of my knowledge.

<i>Shamai M. Carter</i>	301-319-9948	3/18/2026
Contracting Officer	Phone	Date